Using Psychological Characteristics of Situations for Social Situation Comprehension in Support Agents

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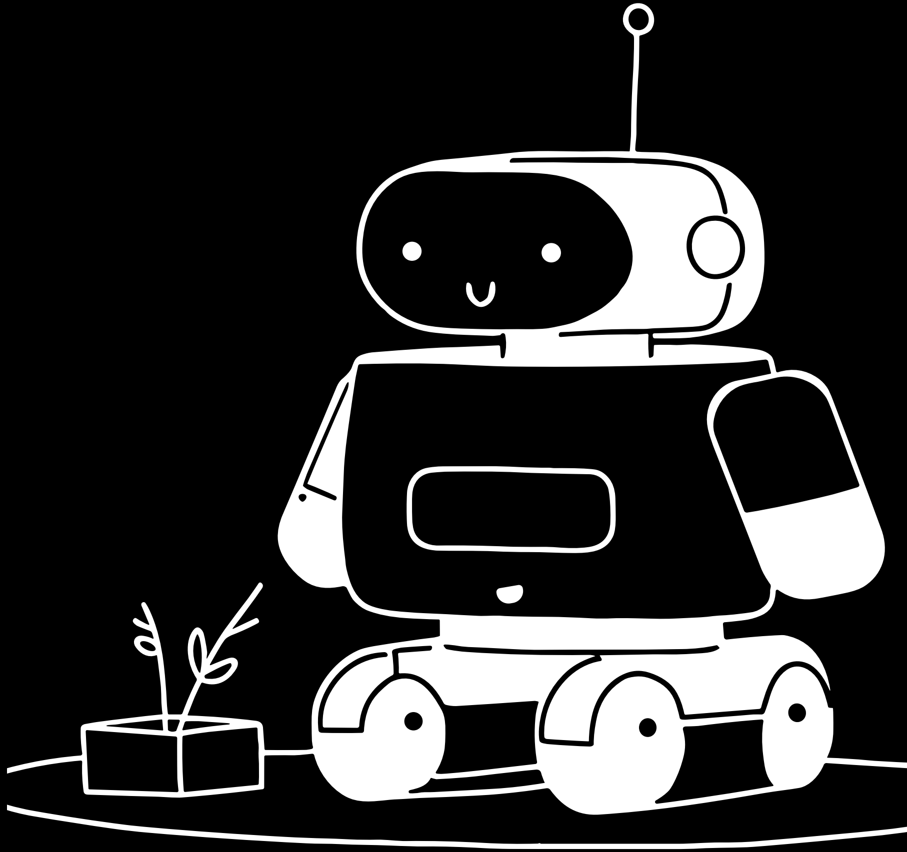
What is a support agent ?

An artificial agent that support people in their daily lives.

Examples : Personal assistants, health coaches, habit formation support agents...



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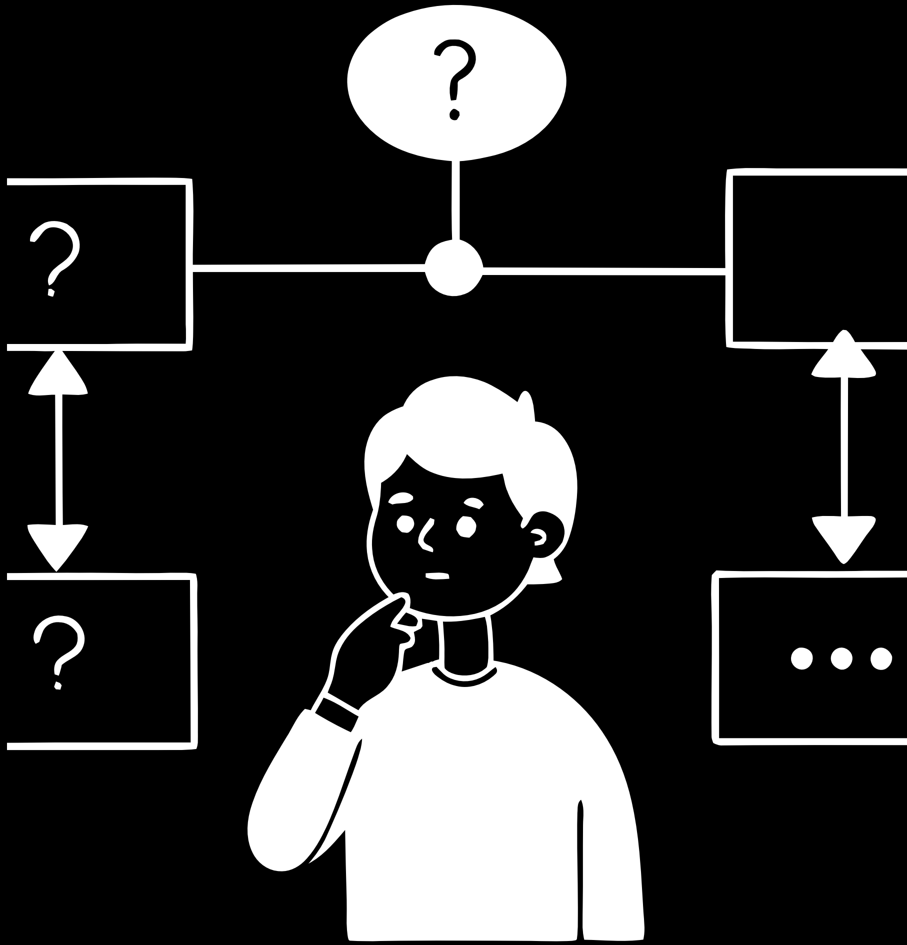
What is the Problem ?

Support agents, to help users, need to take into account not only the user’s characteristics, but also the **social situation of the user**.



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Existing work

Most of the researchers focus on **Personal Characteristics.** They only model the user's internal state (goals, values, emotions).

Limitation: Human behavior is heavily shaped by the environment and situation.

Some have focused on Situation Cues to analyse a situation

- **Setting:** *Work, Home.*
- **Actors:** *Manager, Employee, Friend.*
- **Relationship Characteristics:** *Hierarchy Level, Contact Frequency.*

Social Practices (Dignum) *:* Social practices are seen as ways to act in context: once a practice is identified, people use that to determine what action to follow.

Example : 'Going to work' incorporate the means of transport, timing constraints, weather and traffic conditions, etc.

Simple Feature Set (Kola et al.) *:* Model social situations through a set of social situation features seen from the point of view of the user.

Example : A situation where a manager and an employee are meeting

Setting = Work | Role = Manager | Hierarchy level = Higher.

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Related Work

- Research shows that it is important to determine the meaning of a situation, which is called **social situation comprehension.** (SSC)
- Kola et al. propose a three-level architecture for social situation comprehension that the authors used in this paper.

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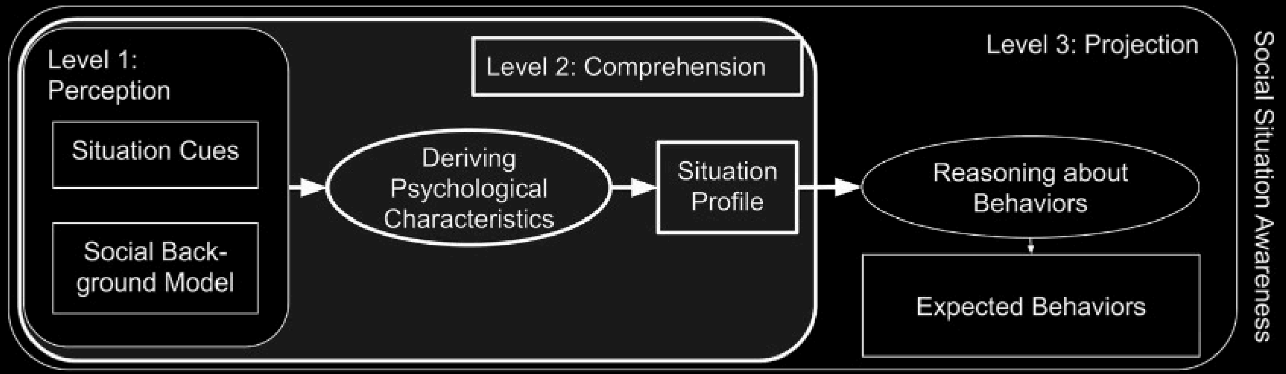

Fig. 1 Simplified version of the three-level architecture for Social Situation Awareness proposed by Kola et al. [28] (emphasis on Level 2 added by us)

Social Situation Awareness Architecture

- **Level 1: Social Situation Perception**
- **Level 2: Social Situation Comprehension. (Paper's focus)**
- **Level 3: Social Situation Projection**

Kola et al. [2]

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What are social situations features (SSF) ?

These are the **objective, observable cues** or facts that define a social situation. They are objective in comparison with PCS.

SSF are the **input** the agent uses to perceive the situation.

Example : Setting (*Work, Home*), Event Frequency, Initiator, Help Dynamic, **Role** of the other person (*Supervisor, Colleague*), **Hierarchy Level** (*Higher, Same*), **Formality Level**...

Limitation: SSF are insufficient because they don't capture the **subjective significance** of the interaction for the user.

Example *:* Two meetings with a "Colleague" (Level 1 features) can have very different impacts (one high **Duty**, one low **Duty**).

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What are the social psychological characteristics of situation (PCS)

They represent the **meaning, traits, or characteristics** that people ascribe to a situation.

As we saw CPS are the **interpretation** derived from SSF, providing the subjective meaning of the situation.

People view situations as real entities and implicitly assign "personality traits" to them, determining how they should respond.

Examples: "Meeting a friend for dinner"

- Low **Duty**
- High **Positivity**
- High **Sociality**

- **"Progress review with supervisor"**
  - High **Duty**
  - High **Intellect**
  - Potential **Adversity**

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DIAMONDS Taxonomy

- **Duty**: Job has to be done; rational thinking called for.
- **Intellect**: Opportunity to demonstrate intellectual capacity.
- **Adversity**: Criticism, blame, or potential threat
- **Mating**: Potential romantic partners present.
- **pOsitivity**: Playful, enjoyable, simple.
- **Negativity**: Stressful, frustrating, anxiety-inducing.
- **Deception**: Someone might be deceitful, causing hostility.
- **Sociality**: Social interaction possible; personal relationships relevant. [2]

Example : Resolve meeting priority conflict between highly **Duty**-driven meeting vs. a low **Duty** but high **Positivity** meeting.

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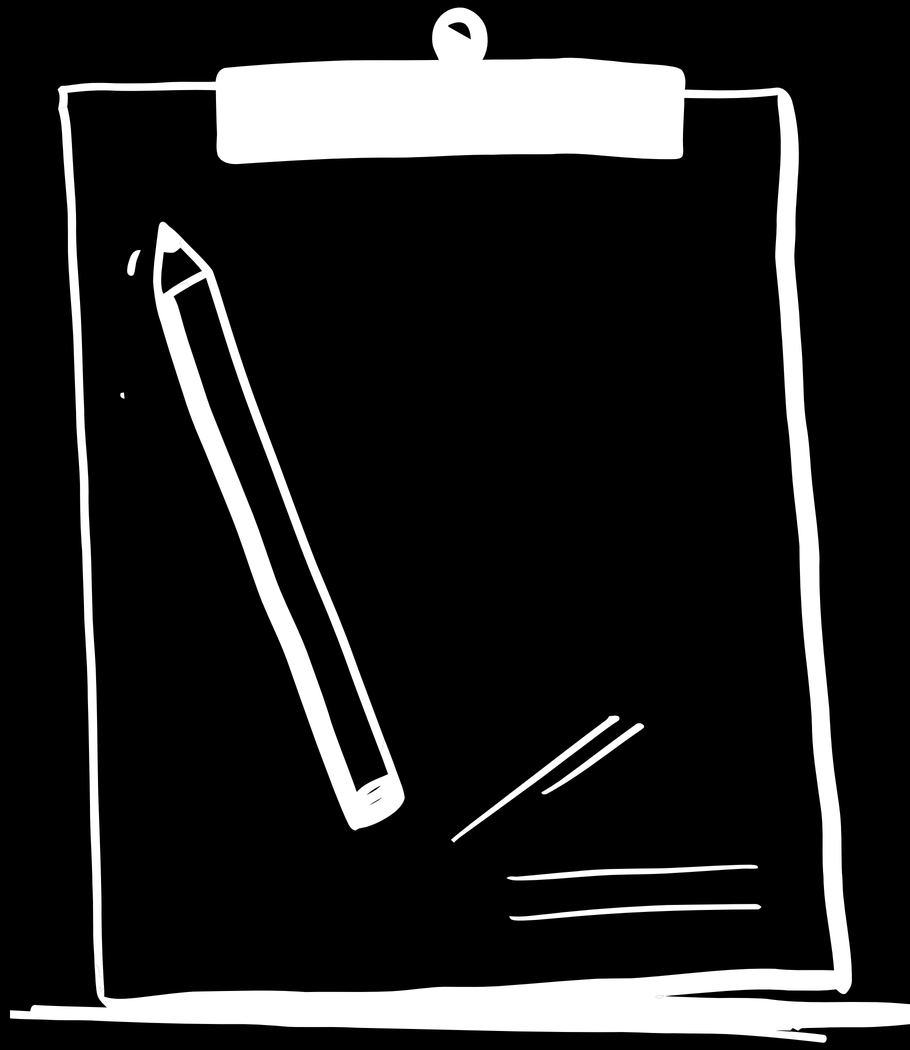
The Methodology

This taxonomy has several advantages.

- **Covers arbitrary situations**, and offers **a validated scale** for measuring psychological characteristics.
- The psychological characteristics of a situation correlate both with **the features of that situation** and with the **behavior people exhibit in that situation.**



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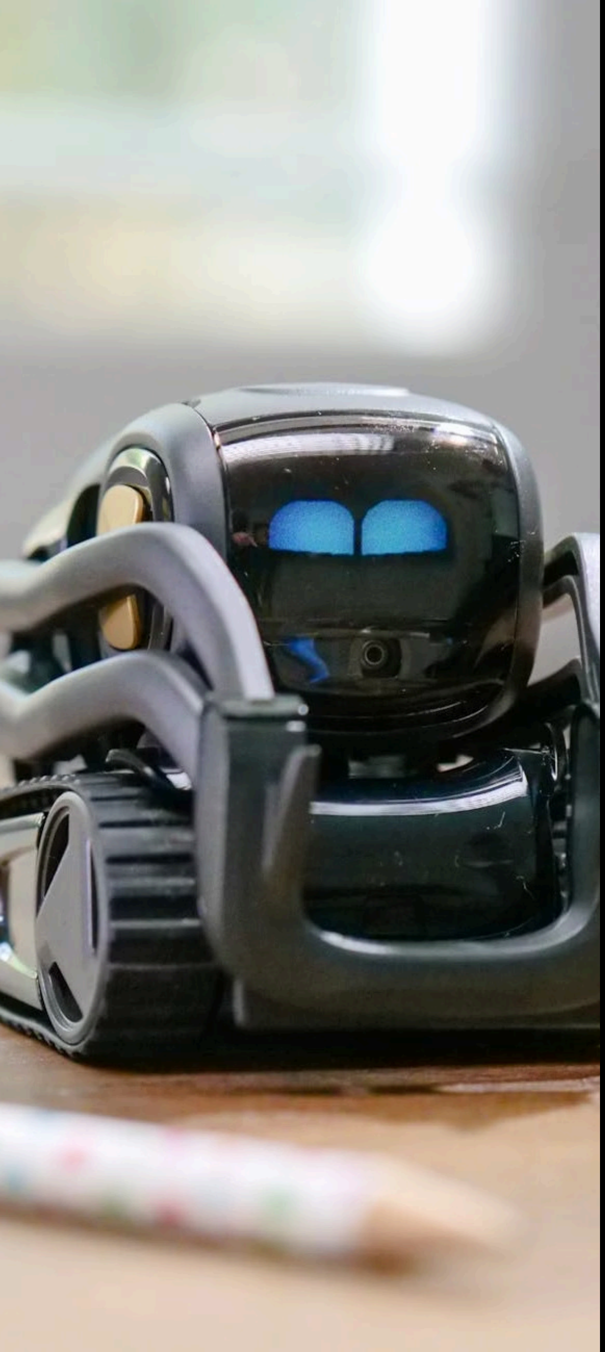

What is AI Explanation (XAI) ?

Definition: XAI refers to an agent's ability to **explain the causes, reasons or internal processes** that led to a decision or suggestion.

Goal *:* To make AI systems more **understandable** and, consequently, more **trustworthy** for users.

- **Local Explanation:** Explains the reasons behind a **specific decision** (Ex : *"The agent suggested Meeting B because..."*). **This is the focus of Study 2.**
- **Global Explanation:** Explains the working of the system in **general**.

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Research Questions

- RH : Using PCS as input in a machine learning model leads to a more accurate prediction of the priority of the social situation than using SSF as input
- RQ1 : Can we use machine learning techniques to predict the PCS (Level 2) using SSF (Level 1) as input?
- RQ2 : Can we use the predicted PCS from RQ1 as input in a machine learning model to predict the priority of a social situation (Level 3) ?
- RQ3 : Can SSF and PCS be used as a basis for explanations that are complete, satisfying, in line with how users reason, and persuasive ?
- RQ4 : When do people prefer PCS in explanations compared to SSF ?

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First study

Goal: To determine if we can we automate the Level 2 step ? (Level 1 → Level 2).

They did a user study to collect **L1 (SSF), L2 (CPS), and L3 (Priority)** on a **7-point Likert scale** for meeting scenarios.

Study 1 is divided into 3 parts :

1. Predictive Power Comparison (L2 → L3 vs. L1 → L3)

Input: L1 or L2 information.

Output: Priority (L3).

Model: Random Forest Regressor (80% Train, 20% Test, cross-validation for tuning).

Result: L2 (CPS) is a **significantly better predictor** of priority than L1 (SSF). They validate the RH.

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2. Automating Comprehension (L1 → L2)

Goal: Predict the 8 DIAMONDS scores (L2) from the SSF (L1).

Models : Decision Tree, XGBoost, **Random Forest**, Multi Layer Perceptron (MLP).

Methodology: 8 distinct models were built (one for each DIAMONDS characteristic) for better accuracy. Models compared against a **Heuristic Model** (predicting the mean).

Result: The **Random Forest model significantly outperforms** the heuristic predictor for **all PCS**

3. Full Pipeline Validation (L1 → Predicted L2 → L3)

Input: The **predicted L2 scores** from the Random Forest models.

Output: Priority (L3).

Result: Confirms the **predictive potential** of the CPS approach.

Critical Note: Highlights the need for **more research** to accurately predict L2, as this is crucial for an overall better L3 prediction.

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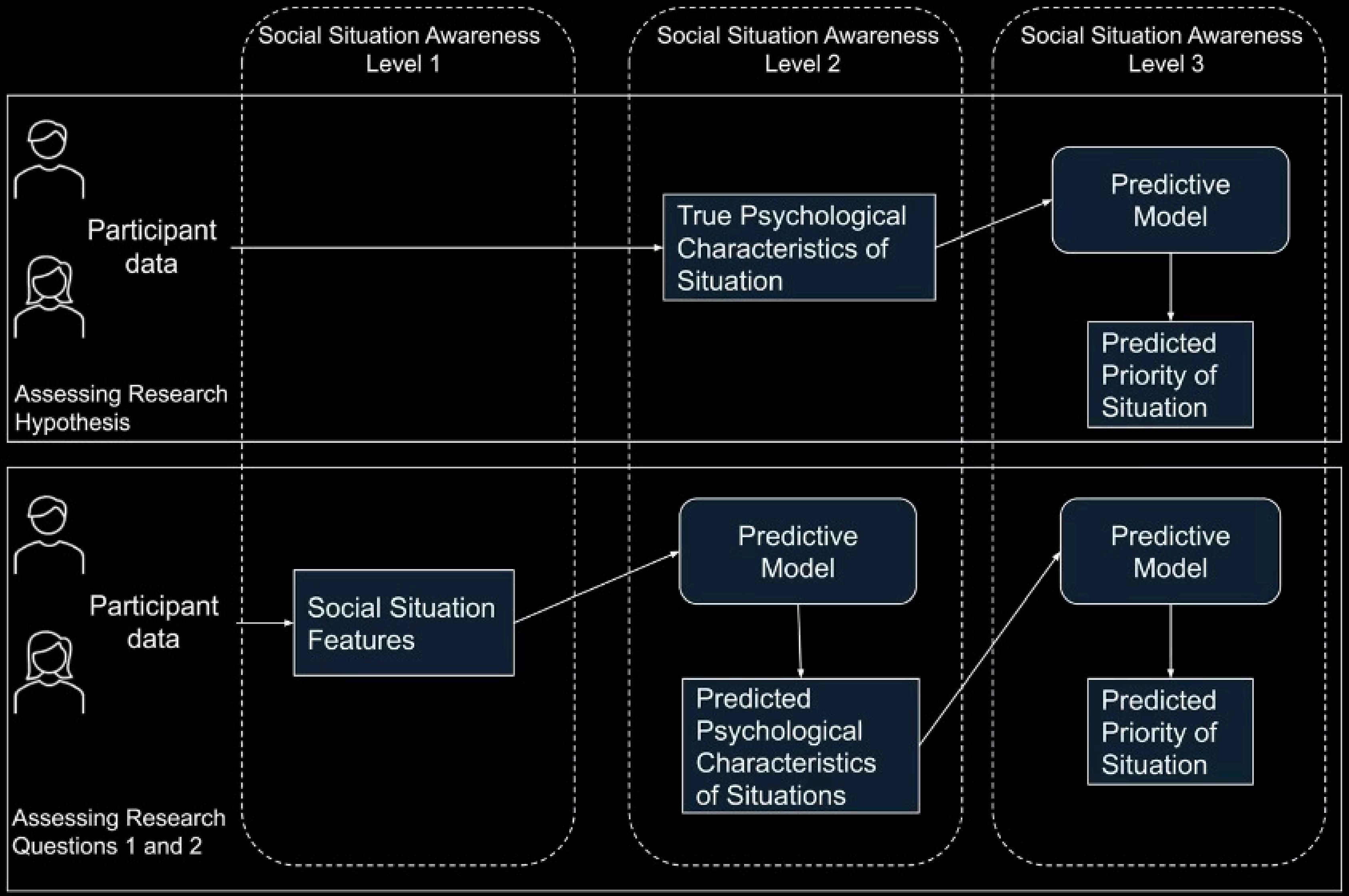

Fig. 2 Conceptualization of Study 1, used to assess the research hypothesis (top part), and Research Questions 1 and 2 (bottom part)

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Second study - Methodology

Goal: Compare the quality/satisfaction of explanations based on L1 vs. L2 information.

Subjects were presented with **pairs of meeting scenarios** and an agent's suggestion with **two types of explanations**:

- **Level 1 Example:** *"...because she is expected to **give help**..."*
- **Level 2 Example:** *"...because it involves a higher level of **Duty**..."*

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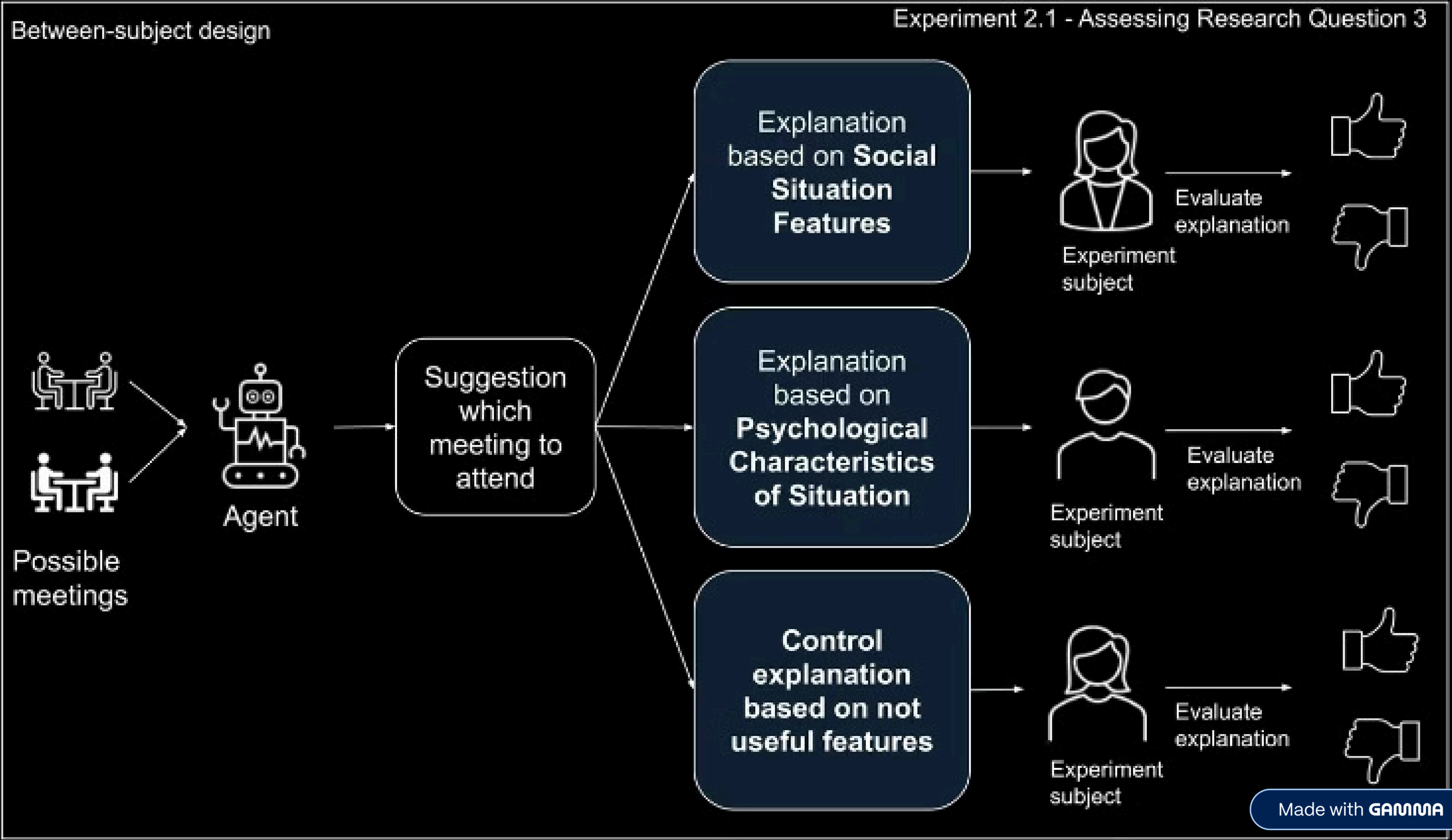

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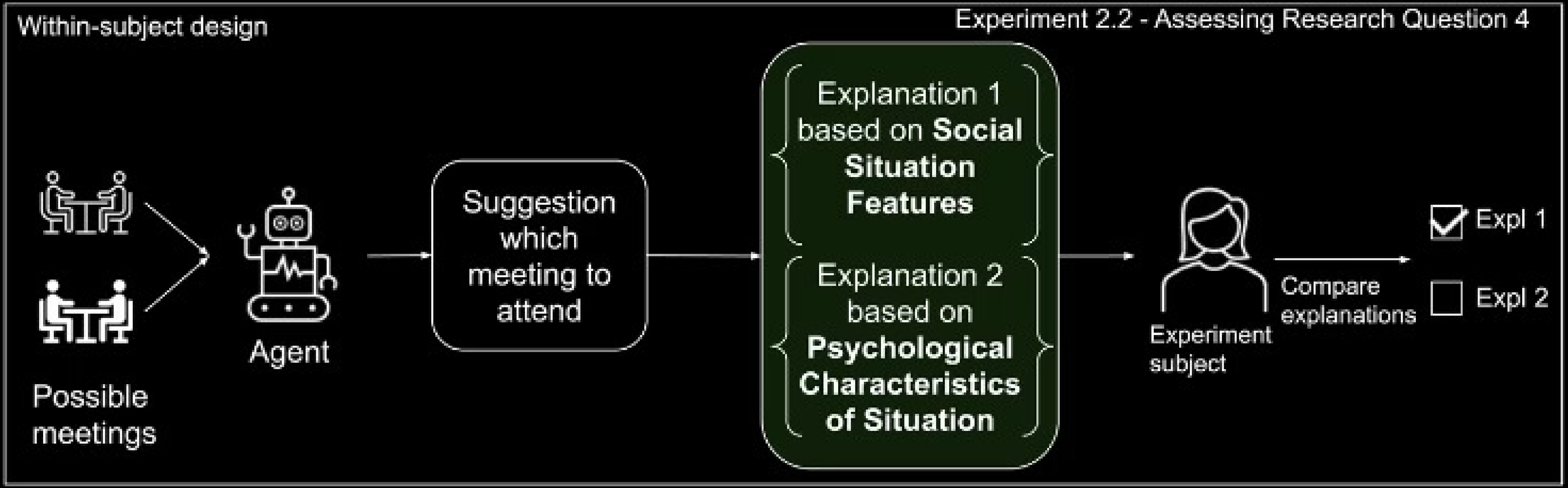

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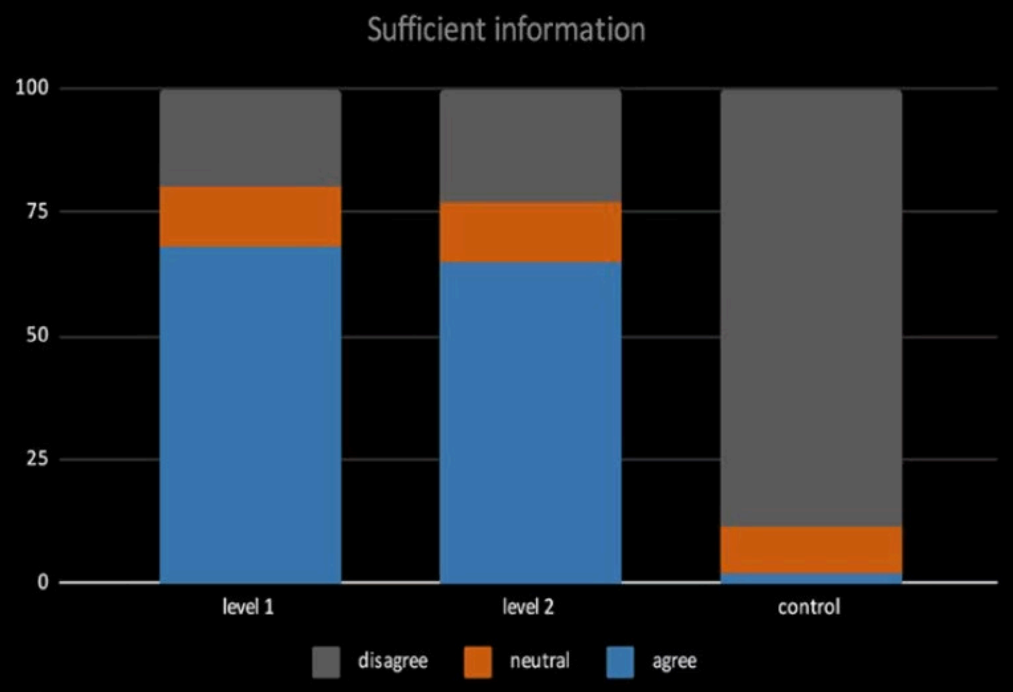

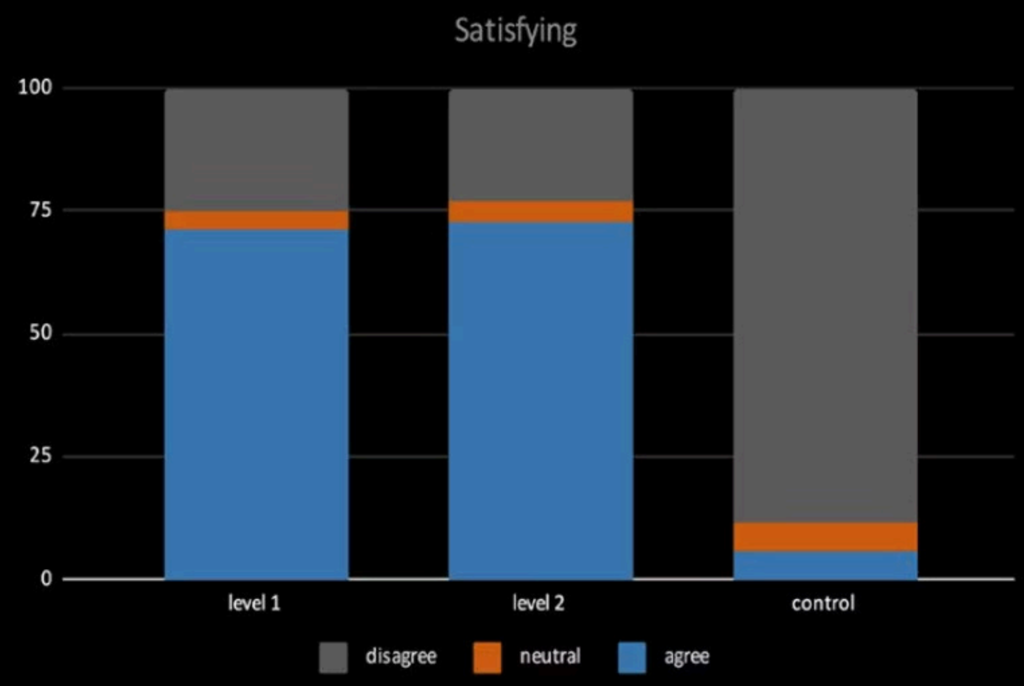

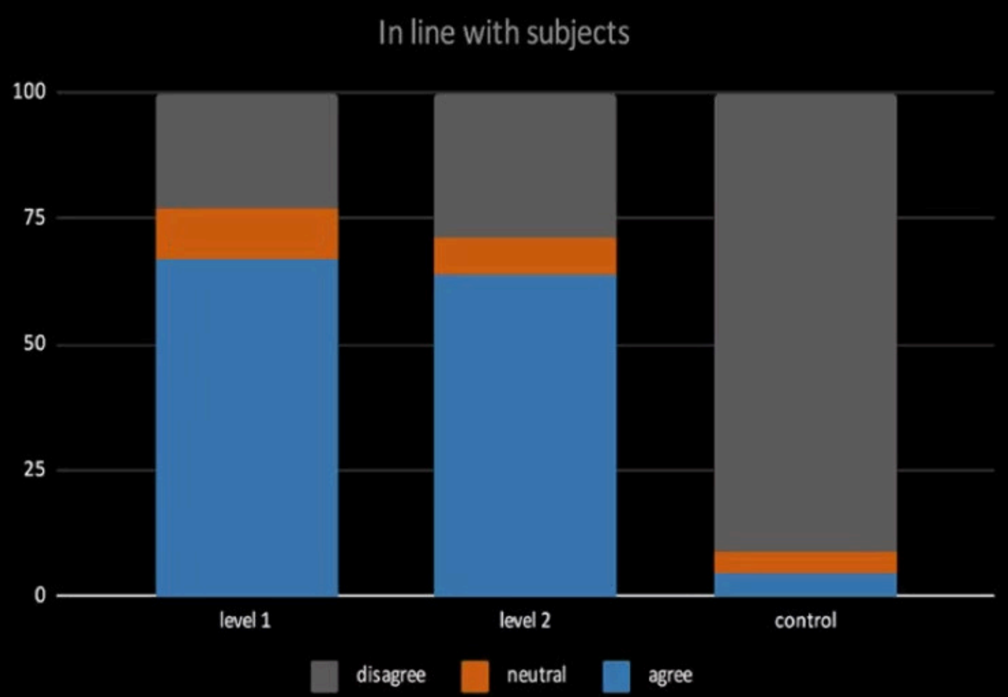

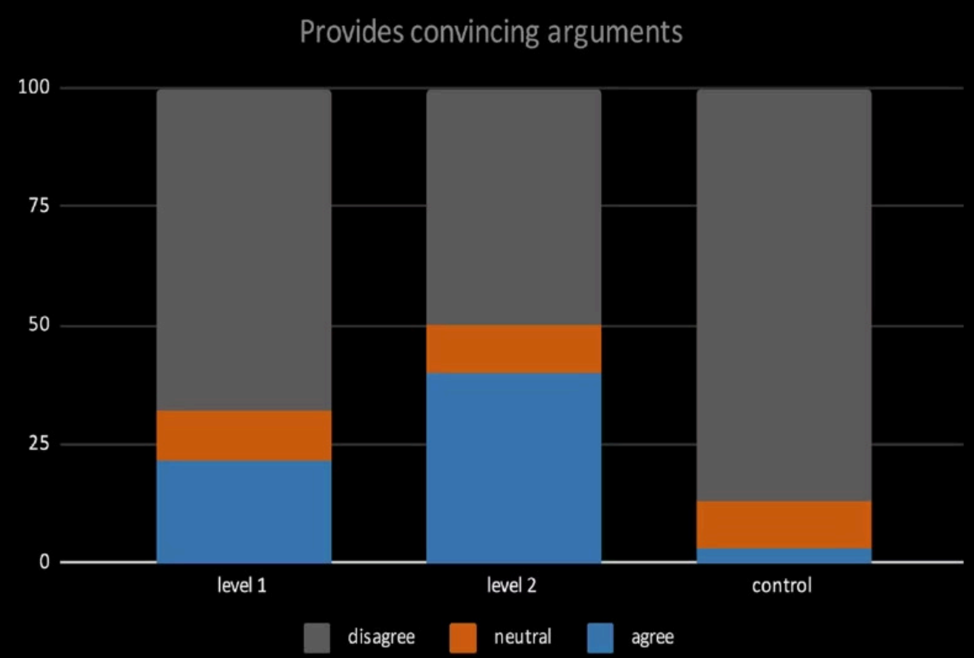

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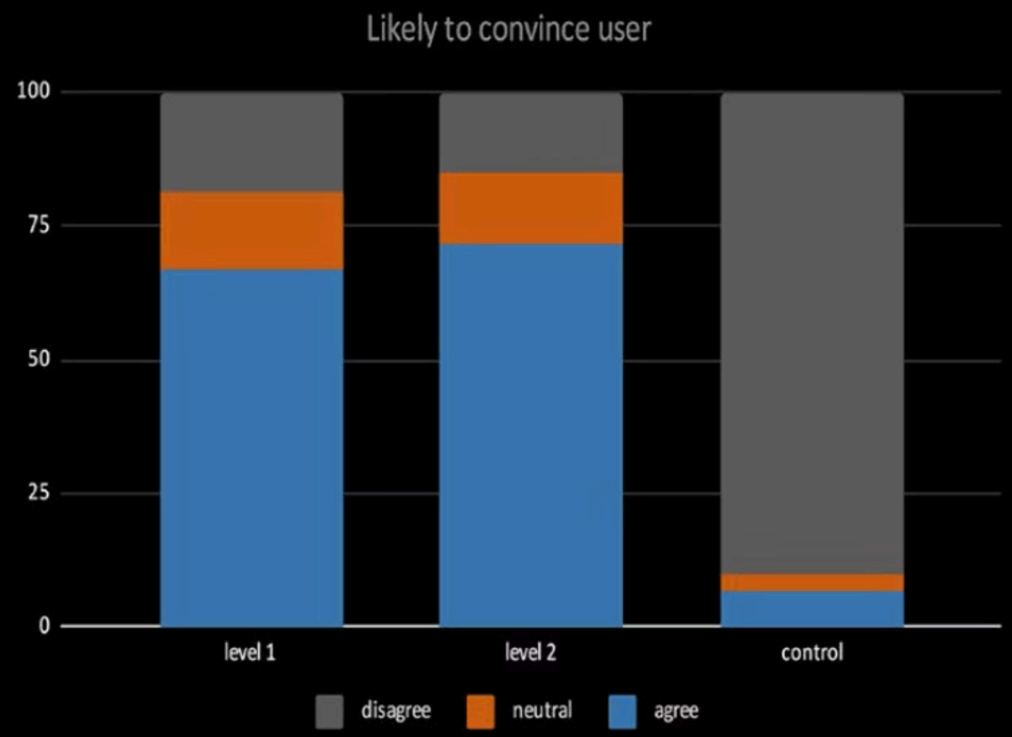

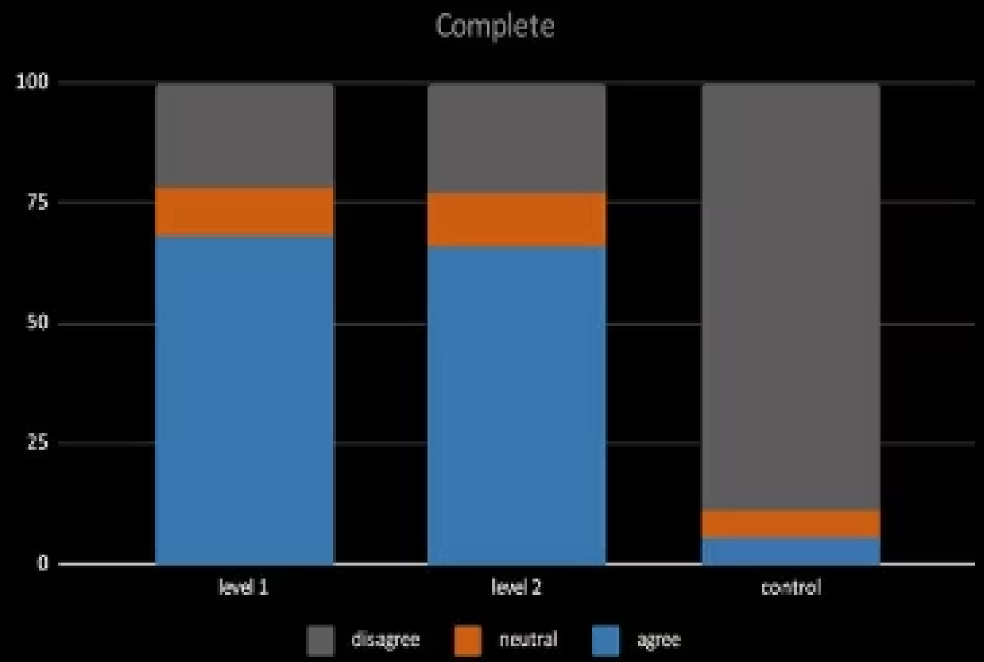

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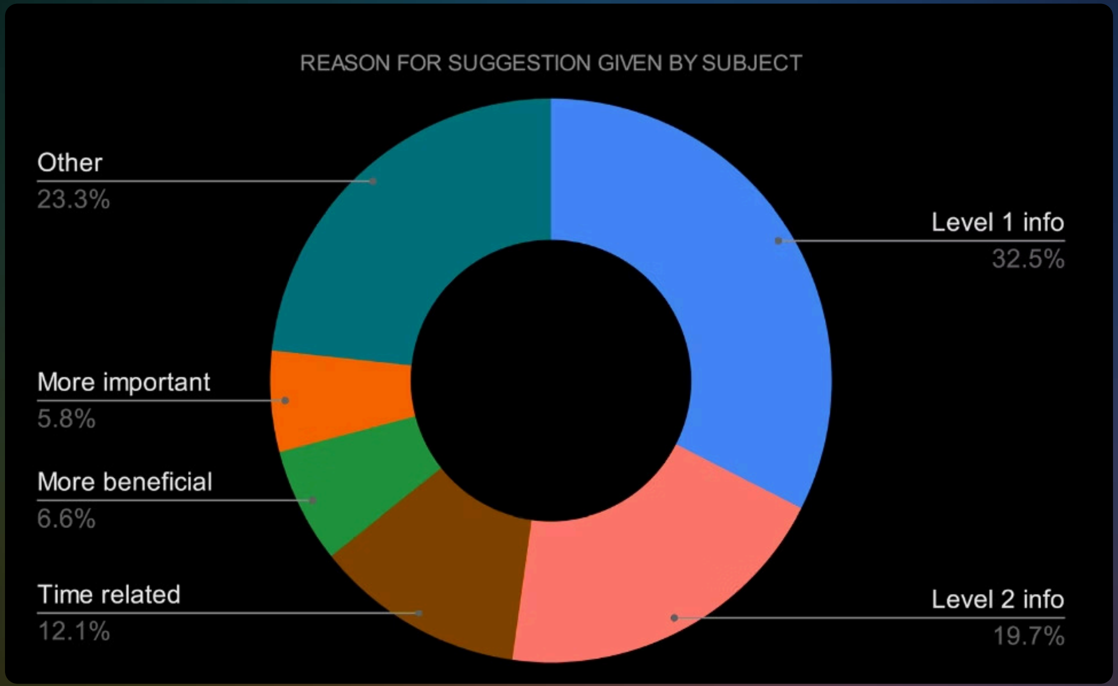

Second study

Results

- Subjects agreed with the agent's suggestion in **73%** of cases.
- Approx. **70%** of subjects found both L1 and L2 explanations to be **complete, satisfying, and sufficient**.

Critical Findings

- Subjects **prefer L2 (CPS) explanations** when **Duty** is the salient (prominent) feature.
- Subjects **prefer L1 (SSF) explanations** when **Negativity** is the salient feature.
- L2 explanations were seen as convincing in **40%** of cases vs. **21.6%** for L1.

Critical Assessment: The low absolute convincingness scores show an **inherent bias**; subjects were implicitly evaluating the *quality of the suggestion itself*, not just the explanation.

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Limits

Element	Point	Detail
Limitation 1 (Data)	Hypothetical Setting	Studies were based on hypothetical scenarios; methods need to be tested in a **real-world prototype** agent.
Limitation 2 (Prediction)	Accuracy of L2	Prediction of CPS (Level 2) needs improvement to maximize overall priority prediction (L3) accuracy.
Ethical Consideration	Misinterpretation Risk	If the agent wrongly infers a high level of **Negativity** for a sensitive situation, it could cause distress or harm to the user.

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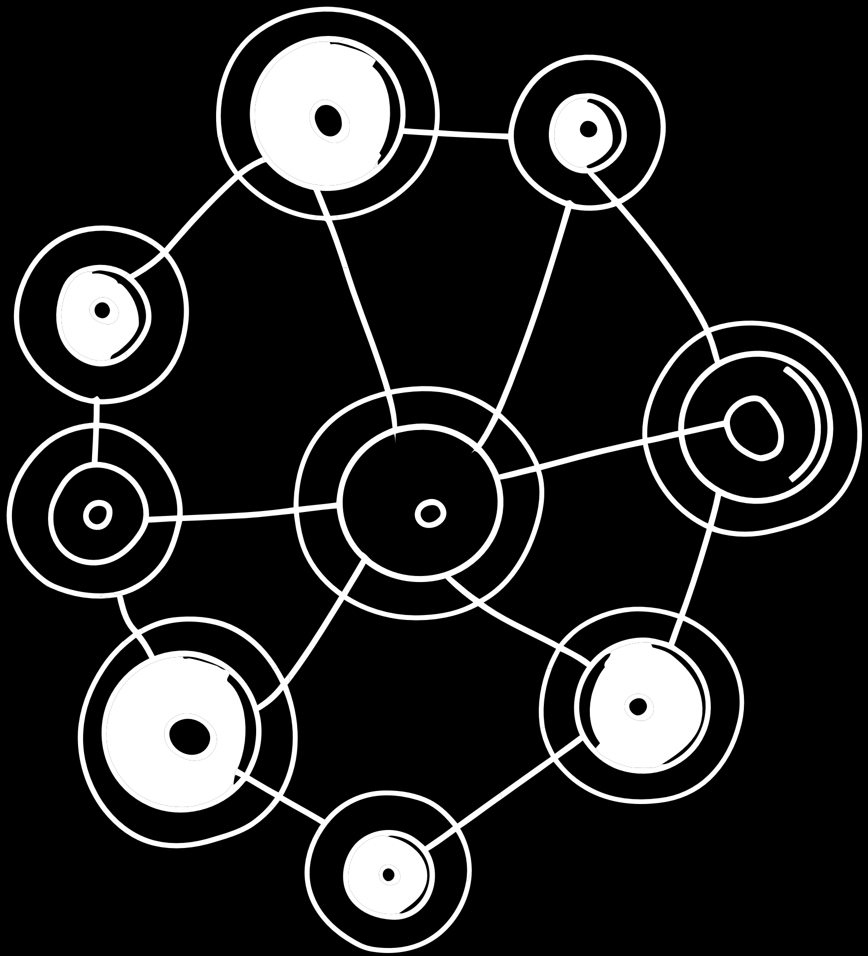

Conclusion

1. This paper validated the use of **Psychological Characteristics of Situations (CPS)** for the Comprehension step (Level 2).
2. Demonstrated that CPS are **superior predictors** of social priority compared to social situation features.
3. Showed that CPS can be **automated** (predicted from Level 1 features).
4. Confirmed that CPS provide a **meaningful and advantageous basis for explanations (XAI)**, especially in "duty-based" contexts.

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My extension proposal : SAGA (Situation Aware Generative Agent)

Implementing the proposed **NLP extraction** of **PCS (DIAMONDS)** from text using a Large Language Model (LLM/GenAI) and/or NLP techniques and integrating this analysis into an **Intelligent Agent System** capable of **Negotiation** with the user.

1. Extract PCS from generated social description with NLP techniques
2. Predict Priority from PCS using different models
3. Build an agent pipeline using **LangGraph** that analyze the text, calculates Priority (Level 3) and provides an explanation for the user. The user can chose to take or not the proposal. If he refuses, the agent tries to negotiate with LLM.
4. Make a comparison with the paper's model

If enough time : Create a user interface with framework like **Streamlit**

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The contributions of the paper

- The authors use psychological characteristics of situations (PCS) for giving meaning to situations, as the basis for **social situation comprehension (SSC)**. **This concept is already used in social psychology**
- **They did two studies**. **The first** shows that PCS can be used as input to predict the priority of social situations, and that PCS can be predicted from the features of a social situation.
- The second shows that PCS can be used as a basis for explanations given to users about the decisions of an agenda management personal assistant agent.

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References

[1] I. Kola, « Using psychological characteristics of situations for social situation comprehension in support agents », 2023.

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